COBRA Invitational 2020

Water Quality B

Team Members: _____

Team Name and Color:_____

Team Number: _____

- This test is made up of four parts:
  - Part 1: Marine and Estuary Ecology
  - Part 2: Coral Reef Macroflora and Fauna Identification
  - Part 3: Water Monitoring and Analysis
  - Part 4: Salinometer Testing
- Teams will be called individually to the station to determine the salinity of the water sample. You must provide your own salinometer and must have goggles.
- For assistance, you may use a calculator and 1 double-sided 8.5x11 page of notes.
- An answer sheet must be used for questions 1-45. Only one answer should be filled in per question. Questions 46-100 must be written directly on the test.
- You have approximately 50 minutes to complete this test. If necessary, correct responses on four pre-selected questions and, if needed, order of completion will be used as tie-breakers.

Time Completed: _____

Part 1 Score: _____

Part 2 Score: _____

Part 3 Score: _____

Part 4 Score: _____

Total Score: _____

TieBreakers: _____

Part I: Marine and Estuary Ecology (30 points)

Match each estuary type from the word bank with the correct example.

1. __D__ Bar-built	A. when the rapid movement of the Earth's crust causes a large piece of land to sink, or subside, producing a depression or basin, filling with ocean water.
2. __E__ Coastal Plain	B. long, narrow valleys with steep sides that are created by advancing glaciers. The glaciers leave deep channels carved into the earth with a shallow barrier near the ocean, which fills with water when the glacier retreats.
3. __C__ Delta	C. form at the mouths of large rivers, when sediments and silt accumulate rather than being washed away by currents or ocean waves. Over time, a complex set of channels, sand barriers and marshes form at the mouth of the river.
4. __B__ Fjord	D. occur when sandbars or barrier islands are built up by ocean waves and currents along coastal areas fed by one or more rivers or streams.
5. __A__ Tectonic	E. As glaciers along the shorelines receded and melted, sea levels rose and inundated low-lying river valleys.

6. What benefit(s) do estuaries provide?

A. Water filtration
B. Habitat protection
C. Provide buffer zones
D. All of the above

7. Coral algae are otherwise known as:

A. Zooxanthellae
B. Fleshy Algae
C. Chrysophyta
D. Pyrrophyta

8. Coral reefs provide habitat to approximately what percent of marine species?

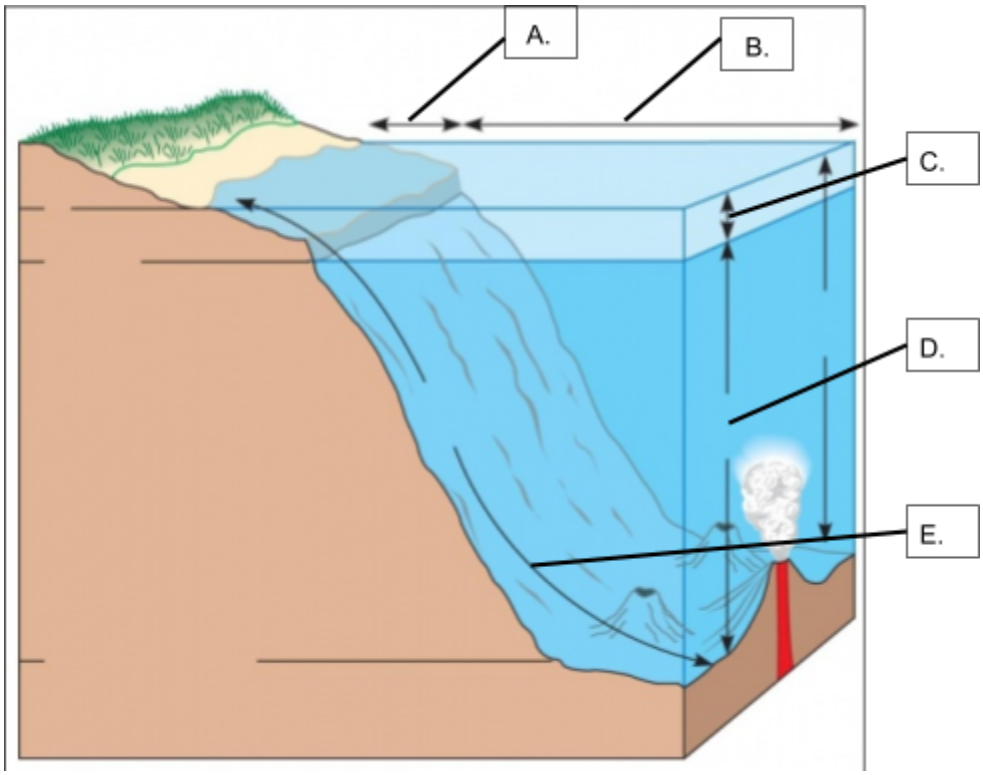
A. 10%
B. 25%
C. 50%
D. 75%

9. Approximately what percent of water on Earth is saltwater?

A. 2%
B. 50%
C. 71%
D. 98%

10. What type of water holds the most oxygen?

A. Salty
B. Fresh
C. Cold
D. Warm



11. Which of the areas on the diagram above indicates the neritic zone? __A_____

12. Which of the areas on the diagram above indicates the aphotic zone? __D____

13. Which of the areas on the diagram above indicates the benthic zone? __E____

14. Most of the water on Earth cycles very rapidly.

A. True B. False

15. The water cycle would not exist without the sun's energy.

A. True B. False

16. The water cycle plays an important role in other nutrient cycles.

A. True B. False

17. Carbon exists in the atmosphere largely as:

A. Calcium Carbonate
B. Carbon Dioxide
C. Carbon Ions
D. Bicarbonate

18. Carbon is converted to carbon dioxide gas through the process of:

A. Photosynthesis
B. Metabolism
C. Weathering
D. Respiration

19. Carbon enters all food webs via:

A. Carnivores
B. Herbivores
C. Autotrophs
D. Heterotrophs

20. Animals rely on prokaryotes in order to convert bacteria into a useable form.

A. True B. False

21. Atmospheric nitrogen is converted into a useable form through a process called:

A. Denitrification
B. Nitrogen fixation
C. Nitrification
D. Ammonification

22. Primarily, where are phosphate compounds found?

A. Plant bodies
B. Sedimentary rock
C. Marine animals
D. Decomposer

23. Fertilizer carried in runoff may cause excessive growth of algae or other microbes that were previously limited by nitrogen or phosphorus in an event known as:

A. Denitrification
B. Hypoxia
C. Industrialization
D. Eutrophication

24. Energy transfers in a food chain are highly efficient, allowing the increase of the chain.

A. True B. False

25. At each trophic level, what happens to a significant amount of energy?

A. It is passed on to the next trophic level.
B. It is released as a waste product.
C. It is given off as heat as a result of cellular respiration.
D. Is is wasted when an organism dies prematurely.

26. Point source pollution is the most common source of decreased water quality.

A. True B. False

27. The picture shows an example of what kind of pollution?



A. Point pollution
B. Nonpoint pollution

28. The most widely used disinfectant for water treatment is:

A. Cyanide
B. Carbon monoxide
C. Dihydrogen monoxide
D. Chlorine

29. Once decontaminated, spreading sludge in the ocean is a viable option.

A. True B. False

30. To create potable water, both physical and chemical processes are used.

A. True B. False

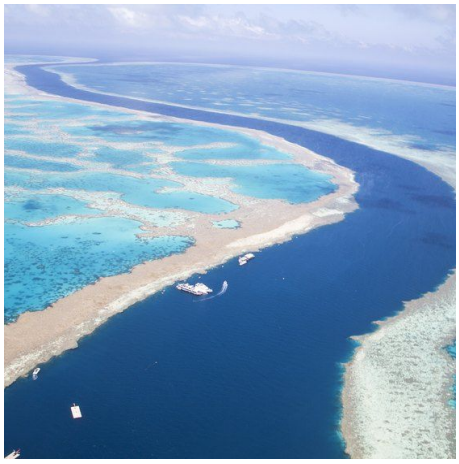
Part II: Coral Reef Macroflora and Fauna Identification (30 points)

31. How long does it take for a barrier reef or atoll to fully form?

A. About 1,000 years
B. 1,001-10,000 years
C. 10,001-100,000 years
D. More than 100,000 years

Identify the following coral reef formations by matching the name with the picture.

A. Fringing B. Barrier C. Atoll

		
32. A	**33. C**	**34. B**

35. Coral bleaching can be caused by:

A. Ocean temperature changes
B. Pollutants
C. Sunlight
D. Any of the above

36. Bleached coral does not contain algae.

A. True B. False

37. When a coral bleaches, it is considered dead.

A. True B. False

38. Soft corals do not produce reefs.

A. True B. False

39. Fleshy algae are important to coral reef ecosystems because:

A. They are the primary producers in many marine food chains.
B. They are resilient to physical and biological disturbances in the reefs.
C. They provide habitat for a variety of coral reef organisms.
D. All of the above

40. Fleshy algae overgrowth can lead to the degradation of coral reefs.

A. True B. False

41. Sponges can only reproduce by budding.

A. True B. False

42. Which species cleans other fish to gather its food?

A. Banded Coral Shrimp
B. Giant Clam
C. Sweet Lips
D. Flamingo Tongue Snail

43. Which species preys on coral polyps?

A. Spong
B. Crown-of-Thorns Starfish
C. Triton
D. Pencil Urchin

44. The most common factor causing the disappearance of species in the Indo-Pacific coral reefs is:

A. Nutrient Pollution
B. Blast Fishing
C. Aquarium Fish Collection
D. Overfishing

45. The chemical used in poison fishing is:

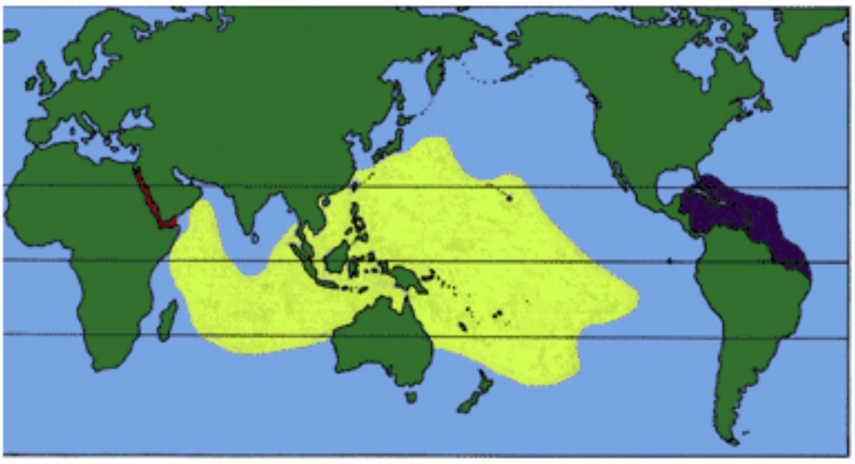
A. DDT
B. Arsenic
C. Cyanide
D. Carbaryl



Stop using the answer sheet and begin writing directly on the test

Use the species on this page to answer the questions on page 9. You may write the entire name or the letter. Some species will be used more than once. Not all species will be used.

 **A.** Barramundi Cod	 **B**. Banded Coral Shrimp	 **C.** Bumphead Parrotfish	 **D.** Butterfly Fish
 **E.** Chinese Mitten Crab	 **F.** Flaming Tongue Snail	 **G.** Giant Clam	 **H.** Gorgonia
 **I.** Humphead Wrasse	 **J.** Lionfish	 **K.** Lobster	 **L.** Moray Eel
 **M.** Nassau Grouper	 **N.** Northern Snakehead	 **O.** Parrotfish	 **P.** Pencil Urchin
 **Q.** Sea Cucumber	 **R.** Snapper	 **S.** Sponge	 **T.** Triton



Coral Reef Regions of the World

Which of the species would **only** be found in the area highlighted in yellow on the map?

46. A-Barramundi Cod	**47.** C-Bumphead Parrotfish
48. G-Giant Clam	**49.** I-Humphead Wrasse
50. Q-Sea Cucumber	

Which of the species would you **only** find in the area highlighted in dark purple on the map?

51. H-Gorgonia	**52.** F-Flaming Tongue Snail
53. M-Nassau Grouper	

Which of the species would people interested in curio collection choose?

54. G-Giant Clam	**55.** F-Flamingo Tongue Snail
56. T-Triton	**57.** P-Pencil Urchin

Which of the species shown are considered invasive species?

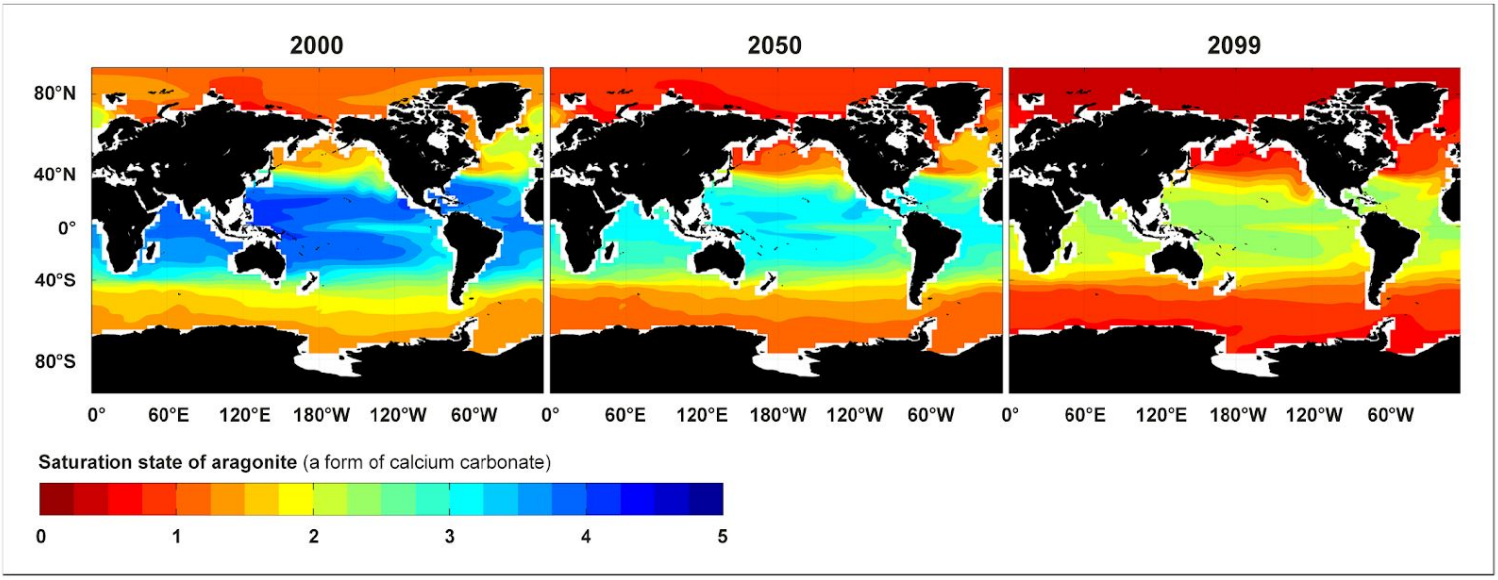
58. J-Lionfish	**59.** E-Chinese Mitten Crab
60. N-Northern Snakehead	

Part III: Water Monitoring and Analysis (30 points)

Fill in the Blanks

Aragonite is the mineralized form of **61.** _____calcium carbonate_____. The amount of aragonite saturation optimal for coral reef growth is **62.** _____4_____ Ω. On Earth, this amount is most commonly found **63.** _____near the equator_____ because **64.** ____the water is warmer/warm water allows it to hold more aragonite_____. The map below shows that the saturation state is **65.** _____decreasing_____. This is most likely specifically due to **66.** _____increased carbon dioxide emissions/ocean acidification_____. If the trend shown on the map continues, organisms such as **67.** _coral/clams/oysters/hard shelled__ to become stressed and even die because **68.** ____they are unable to form hard structures_____ _____.

Figure 2: Modeled Aragonite Saturation State of Surface Oceans, 2000-2099



Source: Katherine Spencer Joyce, adapted from Richard A. Feely, Scott C. Doney, and Sarah R. Cooley ©Woods Hole Oceanographic Institution. | GAO-14-736

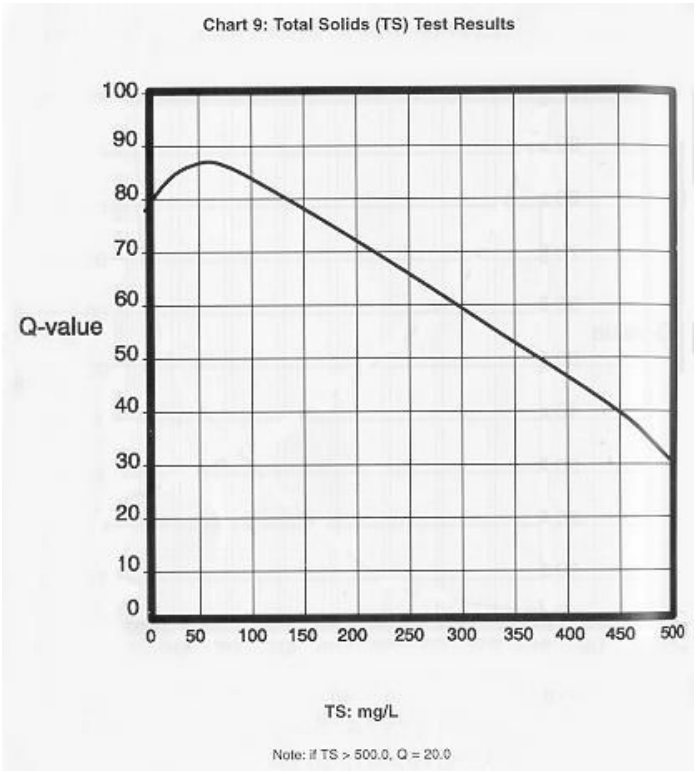
Fill in the Blanks

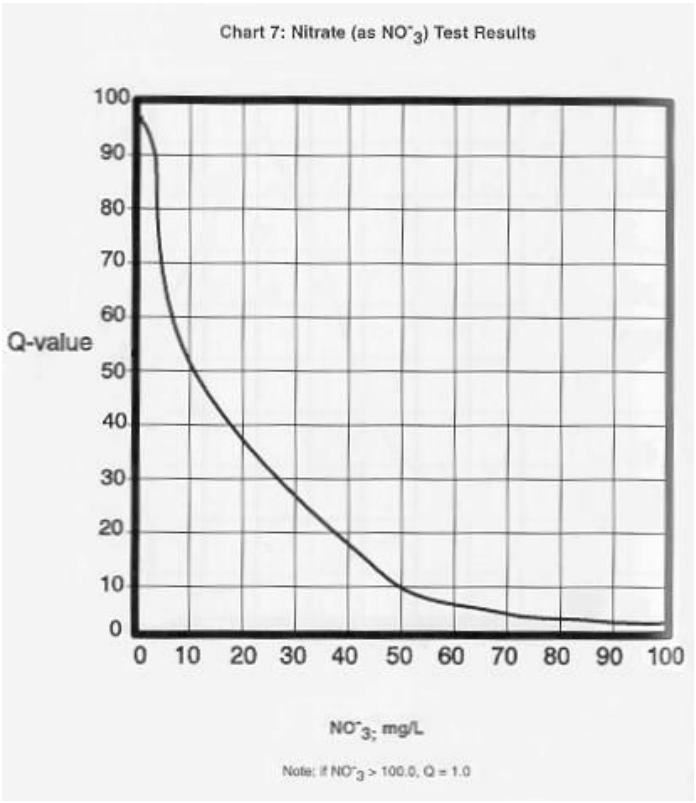
Oceans have a salinity of **69.** ____30-50 ____ppt. Naturally, salt enters waterways from **70.** _contact with rocks containing salt and soluble materials _____. In the ocean, this affects the currents because **71.** salinity helps cause the circulation of the oceans due to the density changes of the water. Excess salt enters waterways as a result of human activity. Too much salt can have negative effects. For example, it can stunt plant growth by preventing the uptake of nutrients. It can affect humans by causing **72.** dehydration/kidney problems It also affects the turbidity of the water by **73.** promoting the accumulation of suspended particles (like clay) into larger chunks, allowing an unnatural amount of sunlight to infiltrate.

Match each water quality parameter with the correct characteristic about the test.

Parameter	Characteristic
74. __G__ Dissolved Oxygen	**A.** Combined total of organic and inorganic substances. This includes anything present in water other than the pure H20
75. __C__ Fecal Coliform	**B.** Does not have a direct effect on aquatic insects or fish. However, excess levels in water can create conditions that make it difficult for aquatic insects or fish to survive.
76. __F_ Biochemical Oxygen Demand	**C.** Disease-causing bacteria, viruses and protozoans may be present in water that has elevated levels of E. coli.
77. __D__ Change in Temperature	**D.** Affects the amount of dissolved oxygen in the water, the rate of photosynthesis by aquatic plants and the sensitivity of organisms to toxic wastes, parasites and disease.
78. __B__ Nitrates	**E.** Measures how far light can penetrate a body of water, may provide shelter and food for pathogens
79. __E__ Turbidity	**F.** The amount of oxygen consumed by bacteria and other microorganisms while they decompose organic matter
80. __A__ Total Solids	**G.** The most important water quality test. Amount is affected by diffusion and aeration, photosynthesis, respiration and decomposition

81. - 84. Complete all missing information in the chart on page 14 using the information on pages 13-14.





Water Quality Rating

91-100	Excellent
71-90	Good
51-70	Medium or average
26-50	Fair
0-25	Poor

Chemical Test Results				
Water Test	**Test Result**	**Q-Value**	**Weighting Factor**	**Water Quality Index**
1. Dissolved Oxygen	80% saturation	86	0.17	14.62
2. Fecal Coliform	0 colonies/100mL	100	0.16	16
3. pH	7.0 units	90	0.11	**81.** 9.9
4. Biochemical Oxygen Demand - BOD	1 mg/L	80	0.11	8.8
5. Change in Temperature	0 deg. Celsius	92	0.1	9.2
6. Total Phosphate - TP	3 mg/L	21	0.1	2.1
7. Nitrates - NO3	2 ppm	**82.** 90	0.1	**83.** 9.0
8. Turbidity	3 JTU	94	0.08	7.52
9. Total Solids - TS	50 mg/L	**84.** 85	0.07	5.95

85. Overall water quality score 83.09

86. Rating: GOOD

87. Explain why weighting factors are necessary. After the Q-value is obtained, it is multiplied by a "weighting factor," based on that test's importance in water quality

88.-90. Based on the data in the **Chemical Test Results** chart, which parameter is the most concerning to the water quality? What is your reasoning for this? What advice would you give to the people responsible for improving the score of this test?

Total Phosphates. It has the lowest Q-value score and its raw score of 3g/mL is above the acceptable score of natural water, which is 0.005 to 0.05 mg/L.

It is important to contain excess agricultural fertilizers, manure, and organic wastes in sewage and industrial effluent

Part IV: Salinometer Testing (10 points)

Did team bring a salinometer? _____

Salinometer Calculation _____8%_____

Calculation Points earned _____

- If the team brought a salinometer, team receives 5 points
- If the team calculation is within + or - 1%, team receives 5 points
- If the team calculation is within + or - 2%, team receives 4 points
- If the team calculation is within + or - 3%, team receives 3 points
- If the team calculation is within + or - 4%, team receives 2 points
- If the team calculation is greater than 4%, team receives 1 point
- If team did not test water salinity, team receives 0 points


DON'T BE THE GREAT
BARRIER.